FACULTY OF ENGINEERING AND TECHNOLOGY

ASSIGNMENT FOR THE
BSC (HONS) IS (BUSINESS ANALYTICS); YEAR 3
BSC (HONS) IS (DATA ANALYTICS); YEAR 3

ACADEMIC SESSION MARCH 2025
IST3244: ADVANCED BUSINESS ANALYTICS

DEADLINE: Week13, 01 August May 2025 Via eLearn

STUDENT NAME: Leow Yan Yi ,Tan Sheng Jing, Chuah Ke En, Peggy Liow En Ki, Isabelle Yau Li Shuang
STUDENT ID: 23020324, 22086243, 21011242, 23020852, 23021165

INSTRUCTIONS TO CANDIDATES

- This group assignment will contribute 70% to your final grade.

Learning Outcome:

LO2: Determine which analytical techniques to apply given the nature of the problem and available data.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 40%.
- Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked zero.

Lecturer's Remark (Use additional sheet if required)

I….....(Name)
……..... std. ID received the assignment and read the comments .....…… (Signature/date)

Academic Honesty Acknowledgement

"I…Leow Yan Yi, Tan Sheng Jing, Chuah Ke En, Peggy Liow En Ki, Isabelle Yau Li Shuang……..... (student name) verify that this paper contains entirely our own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."

LEOW YAN YI TAN SHENG JING CHUAH KE EN PEGGY LIOW EN KI ISABELLE YAU LI SHUANG 31/7/2025
……..... (Student's signature / Date)

Dr Ghulam Murtaza
March 2025

This assignment will allow you to apply your skills in business analytics on data from any of the chosen public dataset which allows you to practice the scientific methods for rigorous testing and documentation. The assignment consists of the applied, "hands-on" development of a prediction model for real-world data and the scientific documentation of your approach. For the first, you will apply the analytics techniques studied in your degree by building a predictive model. For the second, you will document, explain and justify your methodology, experiments, and do a comparison of different model with proper in-depth research justification on the techniques and algorithm differentiation. You would also need to visualize the data using any visualization tools available to present your final results.

A report must be produced, and this would include relevant dashboard, graphs and tables, and it should not exceed 25 pages. You are required to complete the task in a group of 5 students. You need to make use of the scientific literature and conduct and document your experiments with the data. Document down all the stages of modelling process, gathering, cleaning and preprocessing and model selection.
The tasks included can as follows:

1. Select dataset related to business related tasks. For example, sales, marketing, related datasets.
2. Visualize dataset for understanding using any graphical tool for example python graphs or PowerBI.
3. Highlight the data inconsistences, for example missing values, duplicate rows or columns, wrong formats, outliers and etc..
4. Perform data cleaning
5. Create any (three to five) types of models like clustering, association rule mining, regression, trees, Neural network, bagging, boosting, stacking or nearest neighbour and etc.
6. Show result graphs of various models and select the best model with reasoning
7. Perform results comparative analysis of selected models using various performance evaluation metrics.
8. Perform results comparative analysis to a related published study.
9. Discuss limitations and future work.

Breakdown of Tasks by phases:

Phase 1: **Submit a Proposal**. Explain why you selected this domain research and dataset, list down your problem statements, research questions, research objectives, and the outcome of research.
Due 17th June 2025

Phase 2: **Report writing**

1. Propose and justify your solution covering the following aspects:

a) Which techniques/algorithm approach are appropriate?

b) Why these techniques/ algorithm approaches are appropriate?

c) Any previous solutions that are successful/comparable?

d) What would be your proposed solutions?

2. Evaluate your solution and suggest:

a) Are your solutions better than the previous solutions based on literature reviews?

b) What may be the constraints or limitations?

c) What are alternatives that may reduce or minimize the constraints and limitations.

Due Date: **01st August 2025**

Submission: On e-Learn by group leader (Two files/documents **Proposal** and **Report**).
File name format: **Proposal-GroupLeaderName-GroupNumber** and **Report-GroupLeaderName-GroupNumber**
Format: Essay Report
Configuration: Times New Roman, 1.5 lines spacing font size 12
Length: 12,500 words +/- 10%
Turnitin: not more than 20%
Referencing: APA or IEEE

Marking Criteria

Criteria	Max. Marks
Proposal: - Title, introduction, problem statement, research questions, research objectives and keywords (400 to 500 words) - Dataset selection (Give details of dataset like Assignment #1)	20
Dataset domain knowledge understanding	5
Understanding of Data analysis	7
Research Methodology	7
Methods of modelling	5
Literature Review	5
Comparisons of Model	5
Discussions of new findings	5
Visualization /Dashboard	6
Overall report presentation and details of group members contribution	5
Total	70

Marking Rubrics

Proposal: **Criteria**	**Excellent (10-7)**	**Competent (6-4)**	**Poor (3-0)**
Proposal (20 Marks)	Introduction, Problem statements, research questions, research objectives and datasets details are clear and a draft of a good plan for research is presented.	Introduction, Problem statements, research questions, research objectives and datasets details are clear, but research plan is poor or vice versa.	Introduction, Problem statements, research questions and research objectives, datasets details are confusing and is not tied to research planning.
Total 20 Marks			

Criteria	**Excellent (10-7)**	**Competent (6-4)**	**Poor (3-0)**
Dataset Domain Knowledge understanding (5 Marks)	Deep and logical understanding of dataset domain and explained in detailed manner	Some logical understanding of dataset domain and explained briefly.	Little or no understanding of dataset domain and explanation incomplete to understand.
Understanding Of Data Analysis (7 Marks)	A thorough data analysis and the information is accurate; sources are legitimate; appropriate 'reading' of the situations observed.	A brief data analysis but information is mostly accurate; 'reading' of one situation may be questionable; sources good but not varied enough.	Information is unreliable and/or inaccurate; situations observed don't provide valid data analysis
Visualization Using Any Tool Like PowerBI, Excel or Tabulo (6 Marks)	An effective use of various graphs in any tool to visualize, understand the data trends, Discussed in detail.	A moderate use of few graphs in any tool to visualize, understand the data trends, Discussed in detail in a generic manner.	A little of no use of graphs in any tool to visualize to understand the data trends, Discussed in highly generic manner. No specific details.
Research Methodology (7 Marks)	Documents process, explains ideas well, clear introduction and conclusion, obvious transitions, doesn't use jargon, demonstrates knowledge of key points	Document is coherent for the most part, but missing 1 or 2 important elements	Document lacks coherence and is missing 3 or more important elements

Methods of Modelling (5 Marks)	Multiple categories of models with performance visualization are used with excellent explanation and justification provided to choose models.	A few categories of models with performance visualization are used with brief explanation provided for model selection.	Model are used without categorization with poor performance visualization and explanation is vague for model selection.
Literature Review and References (5 Marks) (Key elements: -Coverage of prior research evidence in Machine Learning (ML) and Classification -Relevancy of literature to the concepts/theories/techniques of ML -Correctness of references cited)	**Coverage**: Exceptionally well written review indicates thorough and extensive research from a wide variety of sources **Relevancy**: Literature is thoroughly discussed/compared with justification of its relevancy to the concepts/theories adopted in the project. **Correctness**: Overall, it is error-free with all sources properly cited and it conforms to the required referencing style.	**Coverage**: Good review indicates research from a variety of sources **Relevancy**: Literature is justly and well suitably relevant to the concepts/theories adopted in the project. **Correctness**: A few errors are present with most sources properly cited, as well as the odd error in the required referencing style.	**Coverage**: Weak review indicates little evidence of research **Relevancy**: Literature is weakly relevant to the concepts/theories adopted in the project. **Correctness**: Numerous errors are present with insufficient number of sources cited, wrong and / or missing sources, as well as errors in the required referencing style.
Comparison of Model (5 Marks) (Results of analysis performed, which also includes visual representation, tabular representation and any other form that you want to present your result.)	An exceptionally well presentation of results that fulfil the project objectives and reflect the correctness of results and justification.	The results adequately presented and sufficiently fulfil the project objectives and reflect the correctness of results and justification.	None or too little description to show how the results fulfil the project objectives and reflect the correctness of results and justification.

Discussion of New Findings (5 Marks)	1) The research process summarized well to cover all areas of objectives. 2) Highlights key findings 3) Many insightful implications are drawn from the data analysis and model results	1) The research process summarized but did cover all areas of objectives. 2) Few highlights key findings 3) Few insightful implications are drawn from the data analysis and model results	Poor report with very little or no coverage of findings to achieve the objectives.
Overall report presentation and details of group members contribution (5 Marks)	Properly formatted using same style for heading, body text and figure are properly labelled and cited with proper article referencing. Work distribution among member is excellent.	Weakly formatted using same style for heading, body text and figure are properly labelled and cited with proper article referencing. An imbalance work distribution among members.	Poorly formatted using same style for heading, body text and figure are properly labelled and cited with proper article referencing. A biased or non-realistic work distribution among members.
Total 50 Marks			

Classification model
XGBoost, Random Forest, Logistic Regression, KNN

1.0 Dataset Domain Knowledge understanding

1.1 Domain Understanding

Financial fraud detection is an increasingly critical topic in the finance and cybersecurity industry in this era of rising digital transaction usages. With the utilisations of various finance transaction methods such as digital banking, online payments, and mobile e-wallers, different kinds of frauds have emerged and expanded. According to the Federal Trade Commission (FTC), consumers in the United States have reported losing over $12.5 billion due to fraud in 2024, making it a 25% increase from the previous year (FTC, 2025). This dramatic rise emphasises the need to develop advanced and data-driven fraud detection systems to mitigate such losses in the near future. Fraud in financial transactions typically includes identity theft, account takeover (ATO), credit or debit card fraud, and money mule fraud (Stankovic, 2025). These activities will often be reflected as sudden changes in a person's transactional behaviour and volume.

Although it is an important issue that needs immediate action, it has numerous challenges that require large efforts to be solved. According to Lindemulder & Kosinski (2024), one of the most significant challenges is the emergence of Generative AI (GenAI) technology. Products produced by GenAI such as fake business documents, voice messages, and deepfake videos have been increasingly convincing and synthetic. These products, wrongly utilised by cybercriminals, are the causes of frauds, as humans and detection algorithms are not able to distinguish its legitimacy. These products complicate detection efforts, thereby enabling more sophisticated scams and frauds to happen. Other than that, it was mentioned that the threat environment constantly evolves, and fraudsters can highly adapt and innovate to new detection technologies. In some cases, fraud groups are funded by multi-national criminal organisations that have access to advanced technology and highly skilled personnel. For example, a China-based group BogusBazaar created over 75,000 fake e-commerce websites, which resulted in nearly $50 million of fraudulent orders and more than 850,000 credit card details being stolen (Fadilpašić, 2024).

1.2 Dataset Description

The dataset published by Ashar is a highly realistic simulation of financial transactions designed for fraud detection research and model development. It has over 100,000 rows of data and 21 features. It captures a broad range of variables that represent behavioural, transaction, geographic, and device-based dimensions of a fraud-prone environment. The target variable of this dataset is the "Fraud_Label". Its binary nature makes it highly suitable for supervised classification. The dataset

supports the training and evaluation of machine learning algorithms, including logistic regression, decision trees, random forest, and neural networks.

The key variables in this dataset are:

1. Transaction_ID & User_ID: Unique identifiers for each transaction and user that facilitate user-level behavioural tracking, thus allowing analysts to observe patterns over time.
2. Transaction_Amount: Captures the transaction size of each transaction, where extreme values may signify fraudulent behaviour.
3. Transaction_Type: Differentiates between transaction channels such as Online, In-Store, or ATM, which each might carry different risk profiles.
4. Timestamp: Captures the data and time of the transaction, it enables the detection of irregular transactional timings such as late night activity.
5. Account_Balance: Captures the user's current account balance before the transaction, which helps identify overdrafts or suspicious depletion of funds
6. Device_Type: Captures the type of device used like mobile or desktop, helps identify anomalies when transactions are done from rarely used devices.
7. Location: Geographical location of the transaction, enables the detection of geographically inconsistent transactions that differ from the user's usual geographical patterns.
8. Merchant_Category: Captures the type of merchant, which allows models to flag transactions in high-risk categories or unusual merchant types for the given user.
9. IP_Address_Flag: Shows whether the IP address was flagged as suspicious
10. Previous_Fraudulent_Activity: Captures the number of past fraudulent activities by the user, this is critical for risk scoring based on behavioral history
11. Daily_Transaction_Count, Avg_Transaction_Amount_7d, Failed_Transaction_Count_7d: Number of transactions made by the user within a day, seven days, and number of failed transactions. This allows models to identify deviations from user behaviour.
12. Card_Type & Card_Age: The type and age of the credit, debit, or prepaid card, this provides metadata on payment instruments.
13. Transaction_Distance: Shows the distance between the user's usual location and the transaction location, helps in highlighting transactions which occurred far from the typical location, which may indicate unauthorised access.
14. Authentication_Method: Captures method of user authentication, which provides insights into the transaction's risk based on the method used and security level applied.

15. Risk_Score: Fraud risk score computed for the transaction, is a pre-assessed indication of the likelihood of fraud which aids in prioritising high-risk transactions.
16. Is_Weekend: Indicates whether the transaction occurs on a weekend, it helps identify irregular transaction patterns.

2.0 Understanding of Data Analysis

	Transaction Amount	Account_ Balance	Daily_Transaction _Count	Avg_Transaction _Amount_7d	Failed_Transaction _Count_7d	Card_Age	Transaction _Distance	Risk_ Score
Mode	50000	50000	50000	50000	50000	50000	50000	0
Mean	99.4110122	50294.06598	7.48524	255.2719244	2.00354	120	2499.16416	0.50156
Standard Deviation	98.6872921	28760.45856	4.039637176	141.3822794	1.414272771	68.9858	1442.01383	0.28777
Minimum	0	500.48	1	10	0	1	0.25	0.0001
Maximum	1174.14	99998.31	14	500	4	239	4999.93	1
Q1	28.6725	25355.425	4	132.0825	1	60	1256.4525	0.254
Median	69.66	50384.43	7	256.085	2	120	2490.785	0.50225
Q3	138.8575	75117.365	11	378.0375	3	180	3746.525	0.74958

Figure 2.0.1 Descriptive Analysis

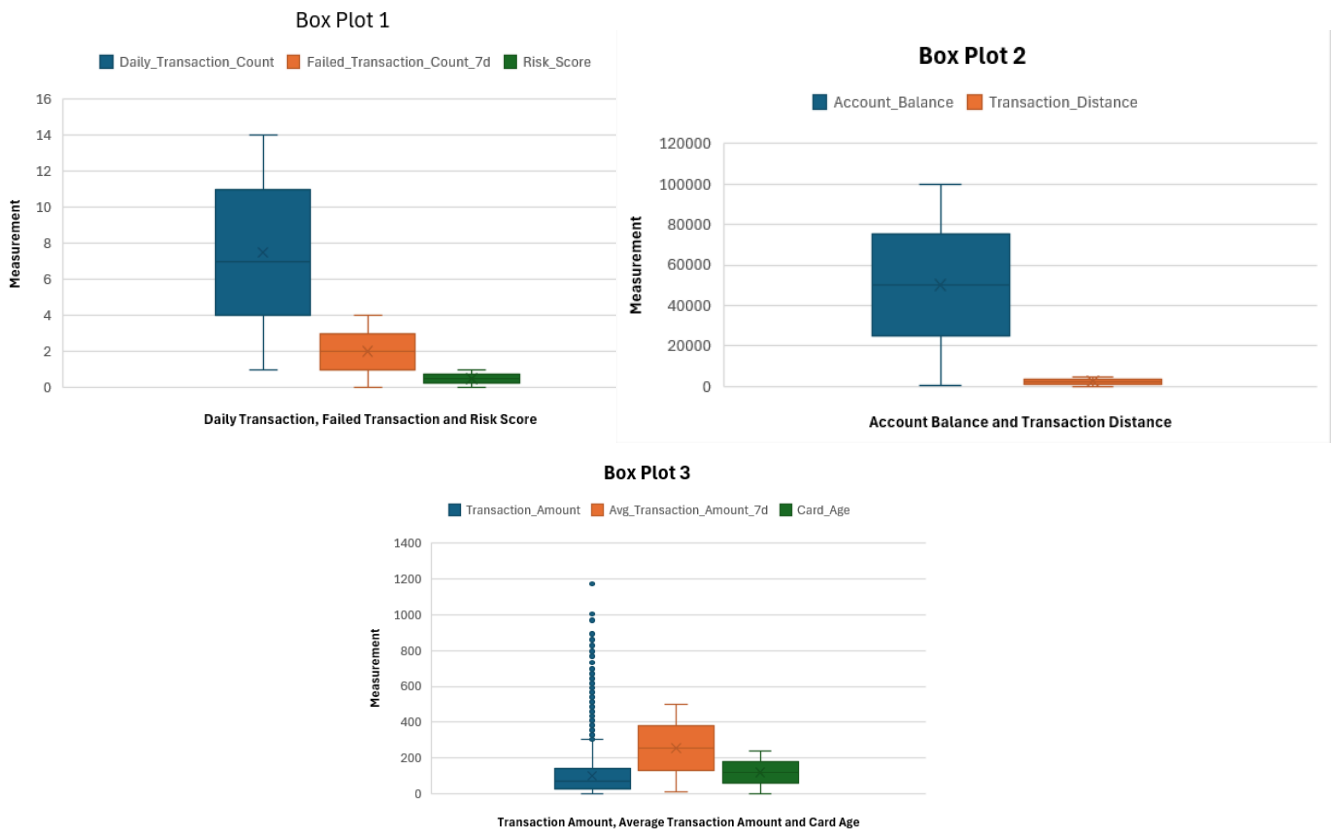

Figure 2.0.2 Outliers Detection

The dataset used was the 'Fraud Detection Transactions Dataset', which contains labeled financial transaction records. The data categorizes each transaction as fraudulent and non-fraudulent based on a variety of numerical and categorical attributes. The objective of this analysis is to comprehend the key features and behaviors in the data, utilizing descriptive statistics and box plots.

Before conducting predictive modeling, it is crucial to understand the underlying patterns, central tendencies, and variations across all variables. This ensures that the analysis is grounded in the real behavior of users and systems involved in financial transactions. The dataset for this analysis of fraud detection is combined with synthetic transaction history, behavioral, temporal, and financial features that are typically associated with real and fraudulent bank transactions. A full exploratory and descriptive analysis on the key numerical and categorical columns was performed, incorporating with summary statistics, which is Table 2.0 and Box Plot 2.0, that may used to identify trends, identify potential areas of risk for classification modeling, and initiate the modeling phase.

First and foremost, the variable Transaction_Amount represents the monetary value of each transaction displays a wide variety Transaction_Amount value, with a minimum of 0 and maximum of 1174.14. The mean transaction amount is 99.41, while having a standard deviation with 98.69 that is relatively large, indicating a variety of transaction amounts. The median is 69.66, which reflects a right-skewed distribution as mentioned in the box plot. This aligns with typical consumer behavior, where small purchases are more common, while outliers may reflect unusually large transactions that warrant further investigation for fraud. The reality financial datasets are expected to display skewness, with most users performing small-to-medium transactions while a minority performs larger-buying activities that could represent either legitimate high-value payments or suspicious activities that need further review.

Besides, the Account_Balance is the account value for the user after each transaction, also displays a low value of 500.48 and a high of 99,998.31, and a mean value of 50,294.07. The standard deviation with 28,760.46 is also relatively high and the IQR, which is Q1 = 25,355.43 to Q3 = 75,117.37 supports the judgement that there is considerable spread in this variable. The box plot suggests that the distribution is right-skewed since there is more distance between the upper whisker and Q3 and the median more towards the lower quartile. Although there are no marked outliers on the box plot, one can infer that given the large scale and resulting range, upper range values may impact fraud risk classifications in different ways, depending on the timing and magnitude of the changes observed in Account_Balance.

Moreover, the Daily_Transaction_Count variable states the amount of transactions a user completes in a day. The minimum value of this variable is 1 and the maximum value is 14, with a mean of 7.49 and a standard deviation of 4.04. From the Box Plot 2.0, this feature has a very slight right skew to it, while the top whisker is longer than the bottom whisker and the median is closer to Q1. The IQR goes from 4 to 11 which tells us that most users typically processed 4-11 transactions

per day. While there do not appear to be any outliers, the shape of the distribution can help with behavioral modeling, as a dramatic increase in transaction activity may not always be a result of legitimate human activity and may be automated or unusual in nature.

Furthermore, the Avg_Transaction_Amount_7d reflects the average transaction amount of a user over the past week. It is valuable in terms of showing transactional patterns. The average is 255.27, resulting in a range of 10 (low) to 500 (high), and a standard deviation of 141.38 gives very good variability when demonstrating weekly transactional patterns producing an IQR from 132.08 to 378.04 to show user variability in weekly activity patterns. The box plot shows mildly right skewness without working very big outliers. The relationship between this variable and single transaction amounts can give insight into abnormal spikes in activity and velocity for sudden change which could also indicate nefarious use.

In contrast, the Failed_Transaction_Count_7d is the number of attempted transactions that failed in the last 7 days. It had a narrower range, from 0 to 4, a mean of 2, and a standard deviation of 1.41. The IQR was quite compact with Q1 = 1 and Q3 = 3, and it was slightly skewed to the right, as presented by the box plot. There were no outliers plotted. In relation to fraud detection, multiple failed attempts is a scare indicator, potentially reflecting unauthorized access or brute force attempts to complete transactions, which makes this feature very worthwhile for downstream analysis.

Other than that, the Card_Age is used to measure the time needed for the user to own the card. It has ranged from 1 to 239, producing a mean of 120 days with a standard deviation of 69 days. Card_Age is otherwise approximately symmetrically distributed with a slight left skew. IQR suggests for cards with their ages 60 to 180 days most cards are quite mature and can show their established use, while also warranting some scrutiny for sudden spikes of activity on newer cards, which is lower Card_Age, suspected of being victim of onboarding fraud, or their newly issued cards misused. Understanding the life span of a customer's card is critical when profiling potential fraud risk, especially since fraudulent behavior may spike during specific stages of card tenure.

In addition, the Transaction_Distance is also an important variable since it is used to estimate the physical and digital location between origin and destination of a transaction. It has a mean of 2499.16 and a high standard deviation of 1442.01. With an IQR that ranges from Q1 = 1256.45, to Q3 = 3746.53, the mid-range is quite broad. Based on the box plot, it reflects a slight right skewness without outliers flagged. Although larger distances may align with digital fraud activity, it is likely going to peak even more loudly if inconsistent with prior actions. Long transaction distances, when

inconsistent with user behavior or device history, could serve as strong red flags for fraudulent activities, especially during weekends or from unknown devices.

Besides, a score that was normalized between 0 and 1 has been provided by the Risk_Score. The higher scores represent higher probability of fraud as inferred from previous transaction history or other external scoring practices. Risk_Score has a mean of 0.50 and standard deviation of 0.29 making the distribution fairly symmetric. The median is 0.5, which is in the figure below in relation to the box also is close to the center and the IQR is between 0.254 and 0.75. It is logical that higher values provided more weight in the prediction of fraud, as this value could have originated from the previous labeling process and could serve as a predictor or validation feature. Given the nature of risk evaluation in fraud detection, it is essential to observe not only the central tendency but also the variability of this score, as it may directly influence predictive model sensitivity.

Although the previous discussion was based on inherently numerical continuous variables, categorical variables are also significant variables in terms of fraud detection. Visualization methods are required for them to be interpreted and understood more specifically. For example, unusual usage of cards by Merchant types, such as luxury good businesses, cryptocurrency exchanges. The unusual spikes in Transaction_Channel types also can suggest there may be a threat or fraud scheme at play or may reflect an unforeseen an shift in patterns of an existing but previously unknown threat. Card type may also affect average spending behavior, transaction volume, and even associated fraud risk, making it a significant attribute when modeling fraudulent patterns.

In addition, while not included directly in the descriptions in Table 2.0, variables such as Transaction_Timestamp, Device_ID, or Location_ID may be part of the full dataset and could indicate temporal or geospatial patterns if incorporated into further analysis. These columns are critical to build user session behavior, detecting velocity attacks, and developing fraud scenarios based on transaction per minute frequency or distance anomalies across locations within a short period.

By combining the use of descriptive statistics and visual analytics, we can derive several important insights. The financial features, including Transaction_Amount, Avg_Transaction_Amount_7d and Account_Balance, exhibit substantial variability are right-skewed, which is typical in many banking datasets, but it could also mean that there are possible low-fraud outliers. The behavioral features, including Daily_Transaction_Count and Failed_Transaction_Count_7d are narrow and low variances, but they are useful when they vary in

actual meaning. The card age and transaction distance are more contextual, but they are critical for understanding overall user behaviors.

Analyzing data distributions with skewness, large IQRs, and possible outliers makes an important point for both modeling and feature engineering, which are normalization and transformation methods, such as log scaled or binned methods are important. Additionally, understanding the distributions is also very important for setting thresholds or cut-offs in classification to detect fraudulent behavior with greater accuracy. For instance, having high Failed_Transaction_Count_7d, low Card_Age, and a large spike in Transaction_Amount all occurring together would demonstrate strong signs of first-party fraud.

In a nutshell, the data set demonstrates realistic financial behavior and plausible outliers, making it a good fit for conducting classification modeling. In this section we used descriptive statistics and box plots to provide the information on measures of central tendency. Each of the attributes we examined is established and adds depth to our understanding of what constitutes suspicious versus normal activity. The analysis of the data done here, will be the basis for developing accurate, interpretable, and robust fraud detection models in later stages of the project. Each analysis conducted not only provides standalone insight into variable behavior but also informs the downstream modeling pipeline by identifying potential predictors of fraud. By combining the descriptive and visual analyses help shape an evidence-based understanding of the data, fulfilling a key role in the fraud detection process.

3.0 Visualization Using PowerBI

3.1 Fraud Count by Fraud Label

Figure 3.1 Fraud Count by Fraud Label

In this section, PowerBI is being used to visualize the data patterns of this Fraud Detection Transactions Dataset. The above Figure 3.1 shows the count of transactions for each fraud category, where 0 represents 'Not Fraud' and 1 represents 'Fraud'. The count of 'Not Fraud' transactions is 33933, which is higher than the count of 'Fraud' transactions, which is 16067. This synthetic dataset is designed with a controlled imbalance to simulate real-world conditions where fraud is less common. In this dataset, fraud makes up about 32% of the data which is higher than that of real-world datasets. In the real world, European Central Bank (2024) mentioned that the fraud in total card payments accounts for 0.015% of the total volume, while fraud in e-money transactions represents 0.012% of all e-payments.

3.2 Fraud Count by Year and Month

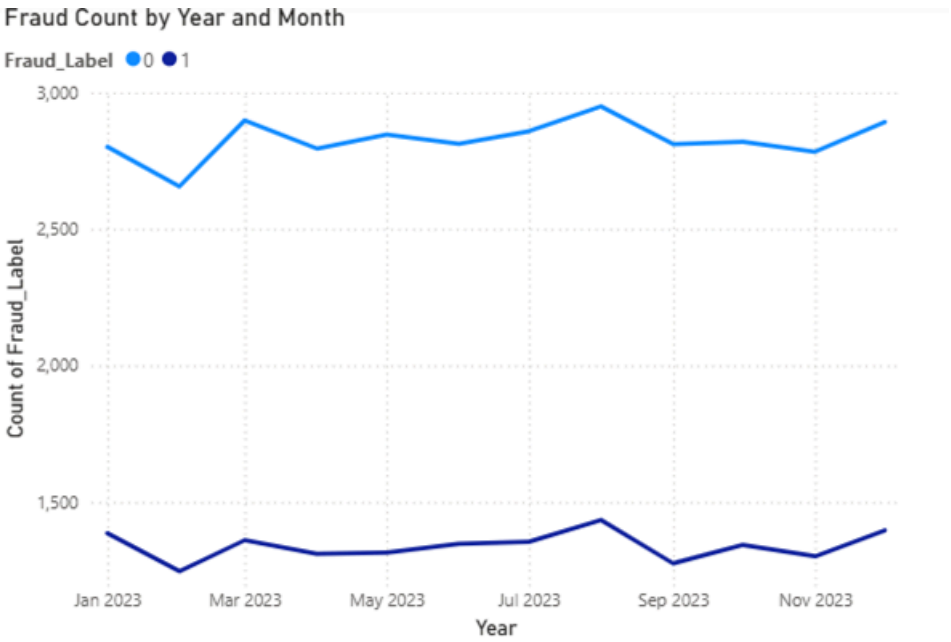

Figure 3.2 Fraud Count by Year and Month

The above Figure 3.2 shows the count of transactions categorized as "Fraud" and "Not Fraud" between January 2023 to November 2023. The X-axis represents date of the transaction whereas Y-axis shows the count of transactions labelled as fraudulent or non-fraudulent. The graph shows a consistent monthly pattern in 2023, with the highest number of fraud cases, recorded 1434 in August 2023 and lowest 1246 in February 2023. The graph doesn't show visible increasing or decreasing trends, which indicates stable fraud rates. The synthetic dataset is balanced for testing model robustness, but it still lacks real-world variability.

3.3 Fraud Count by Transaction Type

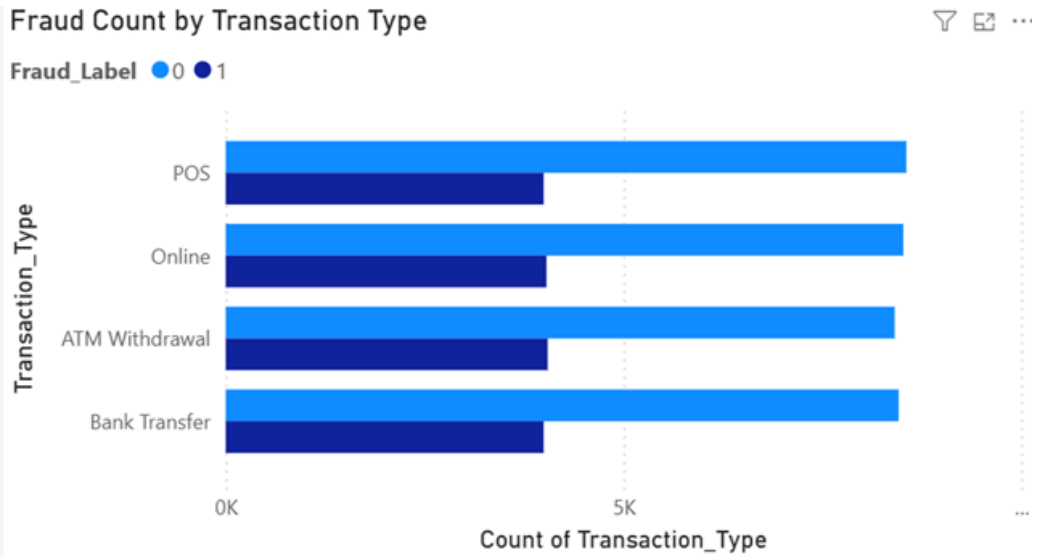

Figure 3.3 Fraud Count by Transaction Type

Figure 3.3 above shows the transaction types against the count of transaction frauds labelled by fraud category. The Y-axis represents the transaction types, which consists of Point of Sales (POS), Online, ATM Withdrawal and Bank Transfer. The X-axis indicates the count of transaction frauds with legend distinguishing between 'Fraud' and 'Not Fraud'. The synthetic data shows even fraud distribution, but the fraud risk is higher in ATM transactions of 4046. The risk is lower in Point of Sales (POS) transactions with a value of 3995. POS sales have lower fraud rates because it uses Chip and PIN technology, which required physical card insertion and PIN entry to make payments, making counterfeit cards harder to use. ATM transactions have higher fraud rates due to hidden skimming devices. These devices are illegally installed on ATM by fraudsters to capture sensitive card information such as card numbers and PINs when users insert their cards.

3.4 Fraud Transactions on Weekends vs Weekdays

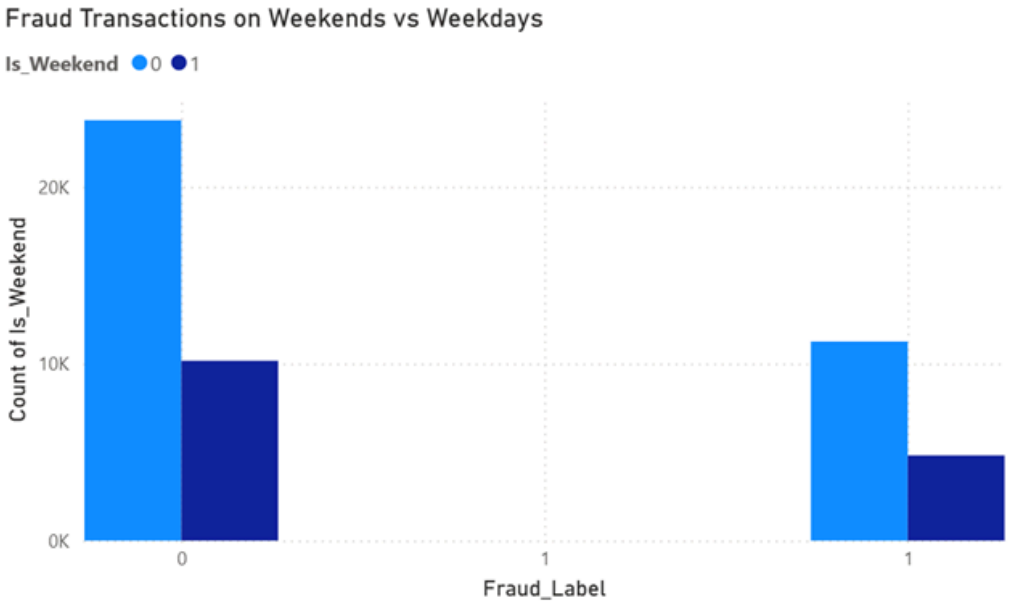

Figure 3.4: Fraud Transactions on Weekends vs Weekdays

Figure 3.4 shows the count of fraud transactions on weekdays and weekends. Transaction occurred on weekdays is represented by 0 and weekends is represented by 1. In Fraud_Label, 0 represents "Not Fraud" and 1 represents "Fraud". Based on the figure, it shows that the number of "Not Fraud" is higher than that of "Fraud". It also shows that fraud that happens on weekdays is higher than that of weekends. This indicates that fraudsters may take advantage of busy weekdays systems where oversight is less tight due to high traffic.

3.5 Fraud Count by Merchant Category

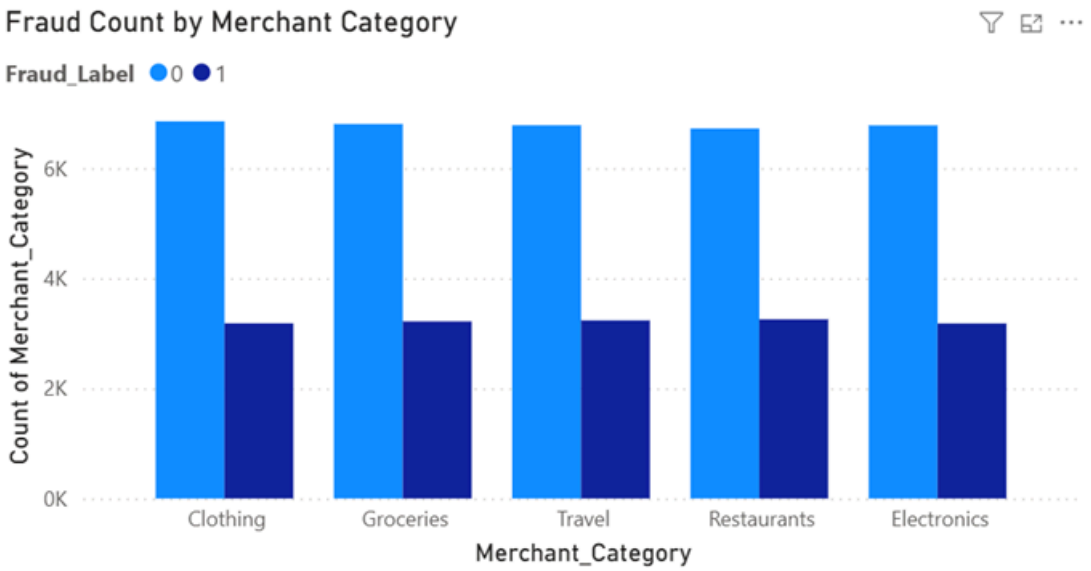

Figure 3.0.5: Fraud Count by Merchant Category

Figure 3.5 shows the distribution of fraudulent and non-fraudulent transaction across different merchant categories. The merchant categories include clothing, groceries, travel, restaurants and electronics. The graph shows that the synthetic data is evenly distributed throughout the merchant categories, with no significant bias toward any category. Travel has the highest number of fraudulent transactions, with a count of 3235, which indicates a higher fraud risk as compared to other merchant categories. This is because travel-related purchases like booking flights are high in value, which makes them more attractive to fraudsters. Clothing has the highest number of non-fraudulent transactions among all merchant categories, with a count of 6852. This is because clothing purchases are more routine and lower in value. Transactions are often performed through trusted and well-secured platforms.

4.0 Research Methodology

The approach taken to address the project's goals and research questions is described in this research methodology. This study aims to predict and comprehend patterns in business-related data by applying a variety of analytical techniques, with a special emphasis on predictive modeling through the use of machine learning algorithms. Data collection, preprocessing, model selection, evaluation, and comparison are all part of the methodology's structured process, which offers a thorough framework for resolving the research issue.

4.1 Research Design

This study uses a quantitative research design, which mostly looks at data analysis and predictive modeling. The goal of the research is to use machine learning algorithms to create models that can accurately predict outcomes based on the data set given. The research design makes sure that the analysis is in line with the research questions and goals and uses the right statistical methods.

4.2 Data Collection

The dataset used for this study comes from the Fraud Detection Transactions Dataset, which is available to the public on Kaggle. There are 50,000 entries and 21 variables in this dataset. Some of the most important variables such as Transaction_Amount, Location, and Avg_Transaction_Amount_7d. The dataset was chosen because it was relevant to the research question, fit with the project's goals, and allowed for the study of important business trends.

The first step in the data collection process was to explore the dataset and perform a preliminary analysis to determine its structure. This required going over the different kinds of variables that were in the dataset, looking at how the data was distributed, and figuring out any possible problems. Gaining a comprehensive grasp of the data's properties and assessing its appropriateness for the study's goals were the objectives.

No additional data transformation or cleaning was required because the dataset was already clean. The dataset was free of duplicates, missing values, and inconsistencies, making it ready for analysis. This made it possible to move smoothly into the phases of analysis and modeling while guaranteeing that the data utilized was precise, trustworthy, and prepared for additional research.

4.3 Data Preprocessing

Data preprocessing is an important step in getting the dataset ready for modeling because it makes sure that the data is in a format that machine learning algorithms can use. First, we changed categorical variables into numbers so that they could be used with machine learning models. This was done by using either one-hot encoding or label encoding, depending on what kind of categorical data it was.

Next, the numerical variables were scaled so that all the features would be the same size. We used standardization methods to do this step, which helped to make all of the numerical features more similar in size. By scaling the data, we made sure that no one feature was more important than the others because of differences in size.

In order to find any extreme values that might skew the results, outlier detection was also done. Using the IQR method, outliers were found and, based on their effect on model performance, either adjusted or eliminated. By doing this, it was made sure that the dataset would always be accurate and that anomalous values would not affect the models' ability to learn from the data.

4.4 Modeling Techniques

Predictive models were developed in this study using a range of machine learning algorithms with the goal of identifying transactions as either fraudulent or not. Based on their suitability for the current classification task and their capacity to recognize patterns in the data, these models were carefully selected. The study used a variety of classification models to approach the problem from various perspectives since fraud detection is a classification problem, with the aim being to predict a binary outcome (fraud or non-fraud).

K-Nearest Neighbors (KNN), a straightforward yet effective classification method, is the first model used in this investigation. KNN classifies data points according to the feature space's nearest neighbors' majority class. This means that transactions with similar attributes such as transaction amount, location, or time, are grouped together for fraud detection, and the model classifies new data points according to how close they are to preexisting ones. KNN works best with smaller, well-structured datasets because it can become computationally costly with larger datasets, even though it is good at identifying patterns where class boundaries are distinct.

Random Forest, an ensemble technique that creates a set of decision trees, is another important model that is employed. By averaging the predictions of several decision trees, Random Forest reduces the risk of overfitting the data, which can happen with a single tree. The ability of

Random Forest to evaluate feature importance and identify intricate non-linear relationships in the data is made possible by this ensemble approach, which is essential for fraud detection. Because of its exceptional ability to handle both numerical and categorical data, Random Forest is incredibly flexible for fraud detection tasks involving a variety of feature types. Additionally, because it can handle high-dimensional data while preserving predictive accuracy, it is especially helpful for big datasets with lots of variables.

Logistic Regression, which is frequently used for binary classification problems, is the third classification model used. Based on its characteristics, logistic regression models the likelihood that a given transaction belongs to a specific class (fraudulent or non-fraudulent). Despite being a simpler model, logistic regression is very interpretable because its coefficients show how each feature influences the probability of fraud. It functions best when there is roughly a linear relationship between the independent variables and the target variable. Even though it is straightforward, logistic regression is a good starting model for classification tasks and is particularly helpful when interpretability is a crucial consideration.

To enhance the overall model performance, XGBoost (Extreme Gradient Boosting) is employed as an ensemble technique in addition to these individual classification models. XGBoost is a potent boosting algorithm that trains several decision trees one after the other, each of which tries to fix the mistakes of the one before it. When compared to individual models, this iterative process frequently results in better performance by reducing bias and variance. Because it can handle large datasets and imbalanced classes, where fraudulent transactions are significantly less frequent than legitimate ones, XGBoost is especially useful for fraud detection tasks. The model is a very powerful tool for fraud detection because of its capacity to regularize and avoid overfitting, which further guarantees that the model generalizes well to new, unseen data.

4.5 Methods of Modelling

The modeling approaches chosen for this study were chosen on the basis of their applicability to the research questions and the dataset's properties. To solve the issue and raise prediction accuracy, a combination of ensemble and classification techniques was selected. Each model was chosen based on its capacity to manage particular data elements, identify significant patterns, and generate reliable predictions. K-Nearest Neighbors (KNN), Random Forests, Logistic Regression and XGBoost were the four models used.

Finally, K-Nearest Neighbors (KNN) was chosen due to its ease of use and efficiency in classification tasks. KNN is a lazy learning algorithm that is non-parametric and uses the majority class of its nearest neighbors to classify data points. This approach is perfect for issues like customer segmentation or pattern recognition because it works especially well when the decision boundaries between classes are non-linear. One of KNN's strengths is its flexibility, which allows it to adjust to the data without making firm assumptions about the distribution of the underlying data. Although KNN can be computationally costly for large datasets, it works best in small, well-structured datasets where it is simpler to capture the relationships between features through proximity.

Random Forests were chosen for this study because of their great versatility in handling both numerical and categorical variables. Compared to single decision trees, Random Forests, an ensemble approach, reduce variance and overfitting by combining the predictions of several decision trees, improving accuracy. Non-linear relationships in the data can also be captured by Random Forests, which is crucial for issues involving intricate variable interactions. Furthermore, Random Forests offer information on feature importance, which is useful for figuring out which variables have the biggest effects on the target variable. Random Forests are a good option for modeling because of their capacity to manage sizable datasets with high dimensionality and deliver reliable results.

The selection of Logistic Regression was based on its ability to solve binary classification problems, which makes it perfect for predicting outcomes with two possible categories, like detecting fraud or determining whether a customer will churn. A popular model that is renowned for its ease of use, effectiveness, and interpretability is Logistic Regression. It produces probabilities, which can reveal information about how likely a particular outcome is. Furthermore, Logistic Regression is a good place to start for many classification tasks because it works best when the relationship between the dependent and independent variables is roughly linear. It is useful because it is simple to use and can be easily interpreted using the coefficients, especially when we need to know exactly how each feature affects the result.

Finally, the advanced boosting algorithm XGBoost was selected due to its capacity to deliver cutting-edge predictive modeling performance. Large datasets and intricate features are easily handled by XGBoost, which also uses boosting to lessen bias and variance. This approach is especially useful for reducing prediction errors because it iteratively enhances the model by fixing mistakes made by earlier models. High accuracy and computational efficiency are hallmarks of XGBoost, which also has built-in tools to deal with imbalanced data—a problem that can arise

frequently in real-world business datasets. Its regularization strategies also aid in avoiding overfitting, guaranteeing that the model performs well when applied to new data. To guarantee the highest degree of prediction accuracy, XGBoost was added due to its impressive results in a variety of machine learning competitions.

4.6 Model Evaluation and Comparison

To ascertain which model would produce the most accurate and trustworthy predictions, it was crucial to evaluate the models' performance after training. Cross-validation, a comparative analysis of all models, and a range of performance metrics were used in the evaluation process. Finding the model that best fit the dataset and the study objectives was made easier by each of these steps.

5-fold cross-validation was used to lower the risk of overfitting and evaluate the models' generalizability at the start of the evaluation process. Using the cross-validation technique, the dataset is divided into five subsets, each of which is used once for testing and the remaining four for model training. Each subset is used as the test set once during the five iterations of this procedure. By using this approach, the models were assessed on various data subsets, improving our comprehension of their performance and guaranteeing that they would function reliably when used on fresh, untested data.

Following cross-validation, the models were assessed using a range of performance metrics. Accuracy, precision, recall, F1-score, and AUC (Area Under the Curve) were the main metrics for the classification models. While precision and recall allowed for a more thorough understanding of the models' capacity to accurately identify positive cases, accuracy offered a broad indicator of how many predictions were accurate. Recall concentrated on the percentage of real positive cases that the model correctly identified, whereas precision measured the percentage of true positive predictions out of all positive predictions. When balancing false positives and false negatives, the F1-score is a useful metric because it provides a balance between precision and recall. Conversely, AUC evaluated the model's capacity to differentiate between the classes; a higher AUC denoted superior performance in differentiating between favorable and unfavorable outcomes.

A comparative analysis was carried out to evaluate each model's performance in relation to the others after they were evaluated using these metrics. Whether it was accuracy, recall, F1-score, or another pertinent metric, the primary metric of focus for this analysis was selected in accordance with the study objectives. This made it possible to compare the models in great detail. The results

were presented more clearly using visual aids like confusion matrices, ROC (Receiver Operating Characteristic) curves, and precision-recall curves in addition to numerical performance. The ROC and precision-recall curves assisted in evaluating the trade-off between various performance metrics, including sensitivity and specificity, while the confusion matrix offered a visual depiction of true positives, false positives, true negatives, and false negatives.

The model with the best overall performance was ultimately chosen following the completion of the comparative analysis and the evaluation of the metrics. As shown by its performance in cross-validation, this model not only did well on the chosen evaluation metric but also showed a strong capacity to generalize to new data. The selected model was thought to be the most suitable for answering the research questions and offering trustworthy forecasts.

5.0 Literature Review

5.1 Overview of classification model in Fraud Detection

Classification model is one of the most common types of machine learning model that classifies objects into classes through finding the similarities of object characteristics (Novakovic et al., 2017). The examples of classification model are logistic regression, decision trees, k-nearest neighbors, artificial neural networks and so on. Classification model works well with problems of supervised learning. The target functions are found to be discrete in classification, as the class labels represent qualitative groups, so assigning them numerical values doesn't carry meaningful mathematical interpretation. Thus, the target attribute in classification is considered a categorical variable.

The performance of the classification model can be measured through metrics like accuracy, precision, recall, area under the ROC Curve (AUC), confusion matrix and F1 score (Cichosz, 2011). Accuracy refers to the ratio of correctly predicted outcomes, which includes both true positives and true negative to the total number of cases examined. It is not reliable when the classes are imbalanced. Precision, which is also known as positive predictive value, represents the proportion of predicted positive cases that are accurate. Besides, Recall is defined as the proportion of positive cases which can be correctly identified. The area under the ROC Curve is often used to measure performance. It is a visual representation which shows the trade-off between true positive rate and false positive rate. Moreover, the confusion matrix shows the true positives, false positives, true negative and false negatives, whereas F1 score is defined as a balance between recall and precision (Novakovic et al., 2017).

The study by Shen et al. (2007) focused on applying and comparing the three machine learning classification techniques, which are decision tree, neural networks and logistic regression to detect credit card fraud, especially external fraud. This paper aims to discover how these models can identify fraudulent transaction through analysing real credit card transaction data from banks and propose the framework to select the most effective model. The study found that neural networks has achieved the highest lift in detecting fraud, followed by logistic regression, whereas the decision trees performed the worst among the three algorithms. The study also used lift charts and capture rates to compare the models. Neural networks have captured 29% of the fraud cases, whereas decision trees captured only 39%.

5.2 Classification Models Algorithms used in Literature

5.2.1 XGBoost

XGBoost, which is also known as eXtreme Gradient Boosting, is a tree boosting algorithm that enhances model performance by optimizing the loss function with added regularization to handle sparse data. It also uses a weighted quantile sketch to improve tree learning. Besides, XGBoost provides features like data compression, sharding and cache access patterns to support scalability in tree boosting system (Zhang et al., 2020). Saw et al. (2024) discussed the benefits of using XGBoost in developing fraud detection models, including the use of gradient boosting to build trees sequentially by correcting previous errors. It also includes regularization to prevent overfitting and enhance generalization as well as parameters designed to handle imbalanced datasets, which is an important feature for credit card fraud detection.

A study by Zhang et al. (2020) has applied the XGBoost algorithm as the classification model to build a customer transaction fraud system. The aim of this study is to identify fraudulent patterns and extract meaningful insights from ordinary data. These fraudulent patterns can then be used in future fraud detection through clustering or classification methods. Besides, the study also compared XGBoost with different models like Logistic Regression, SVM and Random Forest and evaluated them using area under roc curve (AUC-ROC score) and accuracy. The findings show that the XGBoost algorithm outperforms the other three algorithms, as it has the highest AUC-ROC score of 0.952 and the highest accuracy of 0.981, whereas logistic regression has the lowest AUC-ROC score of 0.862 and the lowest accuracy of 0.931.

Besides, Liu et al. (2021) has also applied the XGBoost algorithm into their study to build an online transaction fraud detection system. This study has designed two fraud detection algorithms

based on Fully Connected Neural Network and XGBoost to address the problem of difficult fraud detection in network transactions. The study found that XGBoost outperforms the Fully Connected Neural Network as it has faster prediction times. However, Fully Connected Neural Network's input dimensions are dependent on the specific dataset, any changes to the dataset needs immediate adjustments to the network's structure. The training speed of a Neural network is faster as compared to XGBoost because the XGBoost classifier takes a lot of time to tune the parameters. In short, the Fully Connected Neural Network is more suitable for scenarios with high demand in high-speed processing, whereas the XGBoost-based fraud detection algorithm is suitable for situations that require high precision.

5.2.2 Random Forest

Random forest is a supervised machine learning algorithm that leverages ensemble learning, providing reliable estimates of the generalization error and strong resistance to overfitting. One of the advantages of random forest is its stability, introducing new data point affects only an individual tree instead of the entire model. Besides, the random forest is less prone to bias because it depends on multiple decision trees where each independently trained on different subsets of the data (Jonnalagadda et al., 2019). Jonnalagadda et al. (2019) proposed a machine learning model aimed to detect fraudulent credit card activities in online financial transactions, utilizing the Random Forest algorithm to classify fraudulent credit card transactions. The findings of the study show that the random forest algorithm with enhancements has an accuracy value of 99.93%, making it suitable for larger datasets and offering improved performance with increased training data.

Boyko and Mokryk (2021) mentioned that random forest is an ensemble learning method that combines many machine learning models together to produce results that would be difficult to achieve using a single model alone. The authors have implemented the supervised learning methods, which is the random forest algorithm in the development of an algorithm to detect unlawful transactions that were performed without the permission of the credit card owner. Besides, various algorithms like decision trees, random forest and extra trees are compared to evaluate their optimizations and parameter settings to identify the most suitable algorithm to effectively solve the problem. This study found that the random forest algorithm performs the best for implementing a fraud detection system in banking transactions, as it results in fewer false negative misclassifications. However, the extra trees algorithm required the most processing time on the chosen dataset, making it less efficient.

Moreover, Mihali and Nita (2024) mentioned that Random Forest is an effective ensemble learning technique built based on decision trees, recognized for improving prediction accuracy and reducing overfitting. They have applied Random Forest in their study to discover its effectiveness in handling complex data distributions. Besides, this study provides a thorough understanding of the advantages and limitations of various algorithms in the context of credit card fraud detection. The study found that the enhanced random forest algorithm has achieved a higher F1-score, which indicates a better balance between detecting actual frauds and avoiding false alarms. The enhanced Random Forest algorithm also showed better performance in reducing undetected fraudulent transactions as compared to previous studies.

5.2.3 Logistic Regression

Logistic regression is also one of the algorithms commonly used in developing fraud detection models. One of the key advantages of logistic regression is it performs well on small to moderately large datasets, along with its ability to clearly illustrate how different features affect the prediction. However, logistic regression has one disadvantage where it assumes a linear relationship between the features and the probability of fraud, which limits its ability to capture complex patterns. Other models like random forest and neural networks can model intricate, non-linear relationships, although they come with increased complexity and higher computational costs (Gupta et al., 2025).

Wang and Zhao (2022) modified the logistic regression model to enhance its effectiveness in detecting credit card fraud. They evaluated its performance by comparing model scores and the area under the ROC Curve (AUC) against other algorithms like Support Vector Machines, K-nearest Neighbours and Decision Tree Classifiers. The study begins by adopting the SMOTE method, followed by training the logistic regression model and identifying the best parameters using Grid CV search. The best threshold for the model is then selected by comparing the confusion matrices. The findings of the study show that logistic regression model is the most effective because it has the highest accuracy and larger AUC. Besides, the learning curve shows that the model is well-balanced, indicating neither overfitting nor underfitting.

Moreover, a study by Alenzi and Aljehane (2020) implemented the logistic regression to build a classifier for detecting fraud in credit card transactions, aiming to improve detection accuracy by handling common data issues such as imbalance data. The study also carried out comparison between the proposed logistic regression model and other models like support vector machines and voting classifiers. Although logistic regression had the longest processing time due to database

division and training sub-classifiers, it achieved the best accuracy among the other algorithms. The authors think that detection accuracy is more of a concern than computational efficiency.

5.2.4 KNN model

K-Nearest-Neighbor (KNN) is a classification algorithm, which evaluates each unknown object in the test data set by identifying the k nearest examples from the training set. The study by Shrestha and Pokharel (2023) aims to discover the performance of KNN and k-means algorithms in detecting fraudulent credit card transactions. It uses confusion matrices to evaluate the algorithms based on performance metrics like accuracy, recall, precision and execution time. The study found that KNN's accuracy improves with an optimal k and misclassification increases with extreme values of k. On the other hand, K-Means is better in purely grouping fraud cases without mislabelling normal transactions. In short, K-Means provides more accurate data grouping, while KNN offers a simpler approach by relying solely on k nearest neighbour for classification.

Khodabakhshi and Fartash (2016) has implemented the KNN algorithm to detect fraud in credit card transactions within electronic banking systems, focusing on identifying outliers which are often ignored by other models. The study has combined KNN with association rules and feature weighting to improve the fraud detection accuracy. Firstly, the dataset is divided into two categories which are suspicious transactions and normal transactions. The Apriori algorithm was used to extract the frequent patterns from two types of transactions and train the KNN classifier. When the new transaction is made, the KNN model compares it to known normal and suspicious patterns. Then, the transaction is then classified according to the majority of its k-nearest neighbours using both the legal and fraud pattern datasets. The study found that the proposed method has achieved an accuracy rate up to 98.5%, recall 100% and precision 98%.

Moreover, Kaur (2020) have applied KNN into their study. They have proposed a hybrid of KNN and Naïve bayes classifiers to detect credit card fraud. KNN was used as the base classifier to generate initial predictions. Then, these predictions were passed as an input to the Naïve Bayes classifier, which produced the final classification result. The results of the study show that the hybrid of KNN and Naïve Bayes model has outperformed other models like Random Forest, AdaBoost, Voting Classifier and Naïve Bayes alone. This proposed model has achieved up to 99.5% accuracy and lower execution time as compared to other models.

5.2.5 Naive Bayes Algorithm

Naive Bayes is a type of probabilistic classifier based on Bayes' Theorem, which calculates the probability of a class using prior knowledge and assumes that all the features are independent. This method remains popular among academics and researchers in fraud detection due to its straightforward, fast, inexpensive methods for working with small datasets. Vairam et al. (2022) compared the performance of the Naive Bayes method and Voting Classifier in detecting fraudulent credit card transactions. Their studies indicate that while Naive Bayes has performed well on a dataset with a low number of records but many features. However, it needed significant preprocessing and feature extraction to achieve optimal accuracy. The same algorithm also showed improved precision and recall when irrelevant features were removed, highlighting the importance of proper data preprocessing for enhanced performance in classification.

Besides, Borse et al. (2021) employed robust scaling techniques to improve the detection of fraud using the Naive Bayes model. It involved transforming transaction data in order to reduce the effect of outliers and imbalance, which continues to be problematic in fraud datasets. An accuracy of 91% is presented by this model, showing that Naive Bayes combined with suitable preprocessing, produced competitive results faster.

Furthermore, Ngartera et al. (2025) has proposed a hybrid approach by combining Naïve Bayes algorithm with complementary models to detect both email and financial frauds. They also mentioned that hybrid models often outperform standard Naïve Bayes models, especially when the data is complex and unstructured. This study found that while the Naive Bayes algorithm works effectively in structured datasets with clear class boundaries, the naive assumptions it makes could limit its performance in noisy or correlated environments, and it might benefit from ensemble or hybrid frameworks.

5.2.6 Support Vector Machine

Support Vector Machine (SVM) is another commonly used classification algorithm to detect frauds in transactions., which is also used to analyse both classification and regression tasks (Hussain et al., 2021). It effectively addresses classification problems in noisy and complex domains. One of the advantages of SVM is it has a low risk of overfitting. Issues like multiple local minima and the curse of dimensionality also rarely occur while using SVM (Dheepa and Dhanapal, 2012). A study by Dheepa and Dhanapal (2012) has applied SVM to classify the transactions as fraudulent or non-fraudulent based on behavioural patterns like transaction amount, frequency, location and time. Then, the Kernel function, specifically radial basis function kernel is employed to handle non-linear

data separation. Furthermore, the critical SVM parameters are tuned using grid search and cross-validation to maximize accuracy and minimize false alarms. The results of the study show that the proposed model achieves higher fraud detection accuracy and more scalable in handling large volumes of transactions. The metrics like accuracy, fraud catching rate and so on are evaluated using confusion matrix.

A study by Mary et al. (2021) implemented SVM to classify transactions as fraudulent or legitimate based on behavioural patterns. This study aims to address the issue of rising incidence of credit or debit card fraud, as well as the lack of accuracy of fraud detection methods in existing studies. SVM is used to analyse the transaction behaviour like spending patterns, location or frequency. The system will cancel the payment and alerts the user through email if SVM flags a transaction as suspicious. Moreover, the study also compares SVM with other techniques, which are Artificial Neural Network, Hidden Markov Model and Bayesian Network. The study found that the proposed SVM has achieve a medium accuracy of 94.65%, which is the lowest among the compared models, whereas artificial neural network has the highest accuracy of 97.32%.

Hussein et al. (2021) has used three algorithms to detect the number of fraudulent transactions in the datasets, including random forest, SVM and decision trees. These three algorithms are evaluated based on accuracy, precision and recall. This study aims to address the problem of growth in credit card fraud based on online or offline transactions which leads to financial lost. The SVM is implemented in this study by classifying transactions using hyperplane separation, which is suitable for semi-structured data. The study found that SVM works well even when the data structure is unclear or semi-structured. It rarely encounters issues like overfitting and is effective when there is a clear separation between different classes.

5.3 Conclusion

In short, these classification models are important in detecting fraudulent transactions, especially credit and debit card fraud detections. The algorithms that are commonly used by existing studies are logistic regression, random forest XGBoost, K-nearest Neighbors (KNN), Naïve Bayes and Support Vector Machine (SVM). Ensemble-based methods such as XGBoost and Random Forest often outperform individual algorithms, particularly in handling large, imbalanced or complex datasets. Logistic Regression and Naïve Bayes are simpler and efficient in managing smaller or well preprocessed datasets. SVM can manage semi-structured and noisy data with minimal risk of overfitting. Moreover, KNN is straightforward, and it can be sensitive to parameter choices and dataset structure. Most of the findings mentioned that data processing like handling class imbalance

is important to improve the accuracy and reliability of classification models. In the future, fraud detection systems may benefit most from hybrid models which combines strengths of different algorithms for better performance.

6.0 Model Performance

6.1 XG Boost

```
Accuracy              : 0.9994
Precision (pos class) : 0.9984
Recall (Sensitivity)  : 0.9997
F1 score              : 0.9991
Balanced accuracy     : 0.9995
Matthews CorrCoef     : 0.9986
ROC-AUC               : 1.0000
PR-AUC (Avg Precision): 1.0000

Confusion matrix:
 [[6782    5]
 [   1 3212]]

Detailed class report:
               precision    recall  f1-score   support

           0     0.9999    0.9993    0.9996      6787
           1     0.9984    0.9997    0.9991      3213

    accuracy                         0.9994     10000
   macro avg     0.9991    0.9995    0.9993     10000
weighted avg     0.9994    0.9994    0.9994     10000


— 5-fold CV metrics —
Accuracy         : 0.9992 ± 0.0003
Precision        : 0.9981 ± 0.0007
Recall           : 0.9995 ± 0.0003
F1               : 0.9988 ± 0.0004
Roc_auc          : 1.0000 ± 0.0000
Balanced_accuracy: 0.9993 ± 0.0003
```

Figure 6.1: XG Boost's Performance

The XGBoost model in Figure 6.1 demonstrates exceptional performance in detecting fraudulent transactions. Its overall accuracy of 99.94% indicates that nearly all predictions made by the model were correct, covering both fraudulent and legitimate transactions. The precision for the positive class (fraud) is 99.84%, meaning that almost every transaction flagged as fraud by the model was indeed fraudulent, showing a very low false-positive rate. Additionally, the recall (sensitivity) value of 99.97% highlights the model's ability to capture almost all actual fraudulent transactions, minimizing the number of missed fraud cases. The F1-score of 99.91% confirms that the model maintains a strong balance between detecting fraud and avoiding false alarms, which is crucial in fraud detection applications.

The confusion matrix further supports these findings. Out of 10,000 evaluated transactions, the model only misclassified five legitimate transactions as fraudulent (false positives) and missed one fraudulent transaction (false negative). This indicates that the model is highly reliable and effective at distinguishing between fraudulent and non-fraudulent transactions. Both classes show

near-perfect classification, with non-fraud (Class 0) achieving a precision of 99.99% and recall of 99.93%, while fraud (Class 1) achieved a precision of 99.84% and recall of 99.97%.

Cross-validation results confirm the stability and generalizability of the model. Across five folds, the accuracy remained at 99.92% ± 0.03%, with precision, recall, and F1 scores all consistently above 99.8%. The ROC-AUC and PR-AUC scores of 1.0000 further demonstrate that the model can perfectly differentiate between fraud and non-fraud transactions, even when handling class imbalances typical in financial datasets. Additionally, the balanced accuracy of 99.95% ensures that the model performs equally well for both classes, avoiding bias toward the majority class.

6.2 Random Forest

```
— Hold-out metrics —
Accuracy               : 1.0000
Precision (pos class) : 1.0000
Recall (Sensitivity)  : 1.0000
F1 score               : 1.0000
Balanced accuracy      : 1.0000
Matthews CorrCoef      : 1.0000
ROC-AUC                : 1.0000
PR-AUC (Avg Precision): 1.0000

Confusion matrix:
 [[6787    0]
 [   0 3213]]

Detailed class report:
              precision    recall  f1-score   support

           0     1.0000    1.0000    1.0000      6787
           1     1.0000    1.0000    1.0000      3213

    accuracy                         1.0000     10000
   macro avg     1.0000    1.0000    1.0000     10000
weighted avg     1.0000    1.0000    1.0000     10000


— 5-fold CV metrics —
Accuracy         : 1.0000 ± 0.0000
Precision        : 1.0000 ± 0.0000
Recall           : 1.0000 ± 0.0000
F1               : 1.0000 ± 0.0000
Roc_auc          : 1.0000 ± 0.0000
Balanced_accuracy: 1.0000 ± 0.0000
```

Figure 6.2: Random Forest's Performance

The Random Forest model in Figure 6.2 results show perfect classification performance on the dataset. The model achieved an accuracy of 100%, meaning it correctly classified every single transaction as either fraudulent or non-fraudulent. This is further reflected in the precision (100%) and recall (100%) values for the positive class (fraud). Precision being perfect indicates that all transactions predicted as fraud were indeed fraudulent, while recall being perfect indicates that the model successfully identified every fraudulent transaction without missing any. Consequently, the F1-score, which balances precision and recall, is also perfect at 1.0. In addition, the balanced accuracy, Matthews correlation coefficient (MCC), and ROC-AUC scores all being 1.0 confirm that the model's predictions are both highly accurate and unbiased toward any class.

The confusion matrix illustrates this perfect classification clearly. Out of 10,000 transactions, 6,787 legitimate transactions were correctly classified as non-fraudulent (Class 0) and 3,213 fraudulent transactions were correctly classified as fraud (Class 1), with zero false positives and zero false negatives. This means the model neither falsely flagged legitimate transactions as fraud nor missed any actual fraudulent activity. The detailed class report further supports these results, showing 100% precision, recall, and F1-score for both classes, confirming perfect predictive performance for each.

Cross-validation results are consistent with the hold-out test performance, which is an indicator of the model's stability and generalizability. Across five folds, the Random Forest consistently achieved 100% accuracy, precision, recall, F1-score, ROC-AUC, and balanced accuracy without any variation (± 0.0000). Such stability across multiple data splits suggests that the model is highly robust and not prone to overfitting on a particular subset of data.

6.3 Logistic Regression

```
— Hold-out metrics —
Accuracy              : 0.7881
Precision (pos class) : 0.6355
Recall (Sensitivity)  : 0.7983
F1 score              : 0.7077
Balanced accuracy     : 0.7908
Matthews CorrCoef     : 0.5536
ROC-AUC               : 0.8900
PR-AUC (Avg Precision): 0.8079

Confusion matrix:
 [[5316 1471]
 [ 648 2565]]

Class report:
              precision    recall  f1-score   support

           0     0.8913    0.7833    0.8338      6787
           1     0.6355    0.7983    0.7077      3213

    accuracy                         0.7881     10000
   macro avg     0.7634    0.7908    0.7708     10000
weighted avg     0.8092    0.7881    0.7933     10000


— 5-fold CV metrics —
Accuracy         : 0.8008 ± 0.0036
Precision        : 0.6539 ± 0.0052
Recall           : 0.8072 ± 0.0053
F1               : 0.7225 ± 0.0047
Roc_auc          : 0.8977 ± 0.0012
Balanced_accuracy: 0.8025 ± 0.0038
```

Figure 6.3: Logistic Regression's Performance

The Logistic Regression model in Figure 6.3 demonstrates moderate performance on this fraud detection dataset compared to the tree-based models. It achieved an accuracy of 78.81%, which means it correctly classified nearly four out of five transactions. While this is reasonable, it is significantly lower than the near-perfect accuracy observed with Random Forest and XGBoost, indicating that Logistic Regression struggled to capture the complexity of the dataset. The model's precision for the positive class (fraud) is 63.55%, suggesting that among all transactions flagged as

fraudulent, only about 64% were actually fraud, leaving a considerable number of false positives. On the other hand, the recall (sensitivity) for fraud detection is 79.83%, meaning the model successfully identified around 80% of actual fraudulent transactions but missed 20% of them. The F1-score of 0.7077 reflects this imbalance between precision and recall.

The confusion matrix highlights these challenges. Out of 3,213 fraudulent transactions, the model correctly identified 2,565 (true positives) but failed to detect 648 (false negatives). Similarly, it incorrectly flagged 1,471 legitimate transactions as fraudulent (false positives). These errors indicate that while Logistic Regression captured some fraud patterns, it struggled with a clean separation between fraud and non-fraud, likely due to the complex and non-linear nature of the data. The class-wise performance shows that non-fraudulent transactions (Class 0) had relatively higher precision (0.8913) but lower recall (0.7833), while fraudulent transactions (Class 1) had lower precision (0.6355) despite having better recall (0.7983).

Cross-validation results are consistent with the hold-out test, showing average accuracy of 80.08% with minor variation (± 0.0036), indicating the model is stable but inherently limited. Precision and recall in cross-validation remained similar (0.6539 and 0.8072 respectively), and the ROC-AUC score of 0.8977 suggests reasonable discriminatory power but still lower than tree-based models. The balanced accuracy of 80.25% highlights that the model did not heavily favor one class over the other, although the false positive rate remained relatively high.

6.4 KNN

```
— Hold-out metrics —
Accuracy              : 0.8135
Precision (pos class) : 0.7822
Recall (Sensitivity)  : 0.5814
F1 score              : 0.6670
Balanced accuracy     : 0.7524
Matthews CorrCoef     : 0.5529
ROC-AUC               : 0.8852
PR-AUC (Avg Precision): 0.7789

Confusion matrix:
 [[6267  520]
 [1345 1868]]

Class report:
              precision    recall  f1-score   support

           0     0.8233    0.9234    0.8705      6787
           1     0.7822    0.5814    0.6670      3213

    accuracy                         0.8135     10000
   macro avg     0.8028    0.7524    0.7688     10000
weighted avg     0.8101    0.8135    0.8051     10000


— 5-fold CV metrics —
Accuracy         : 0.8141 ± 0.0014
Precision        : 0.7896 ± 0.0026
Recall           : 0.5747 ± 0.0051
F1               : 0.6652 ± 0.0035
Roc_auc          : 0.8855 ± 0.0033
Balanced_accuracy: 0.7511 ± 0.0023
```

Figure 6.4: Logistic Regression's Performance

The K-Nearest Neighbors (KNN) model in Figure 6.4 shows moderate performance in detecting fraudulent transactions. It achieved an overall accuracy of 81.35%, indicating that roughly four out of five transactions were correctly classified. The model's precision for fraud detection (positive class) is 78.22%, meaning that when KNN flagged a transaction as fraudulent, about 78% of those predictions were correct. However, its recall (sensitivity) for fraud is relatively low at 58.14%, showing that the model failed to identify more than 40% of actual fraudulent transactions. This imbalance results in a relatively modest F1-score of 0.6670, which highlights challenges in maintaining both high precision and recall simultaneously.

The confusion matrix illustrates these challenges further. Out of 3,213 actual fraudulent transactions, KNN correctly identified 1,868 frauds (true positives) but missed 1,345 frauds (false negatives). Additionally, it incorrectly classified 520 legitimate transactions as fraudulent (false positives). On the other hand, non-fraudulent transactions (Class 0) performed better, with a precision of 82.33% and a high recall of 92.34%, indicating that the model was more effective in identifying legitimate transactions than fraudulent ones. In contrast, fraudulent transactions (Class 1) had lower precision at 78.22% and significantly lower recall at 58.14%, showing that many fraudulent cases were either missed entirely or misclassified, which limits the model's reliability in detecting fraud.

Cross-validation results confirm this trend and indicate the model's stability, with an average accuracy of 81.41% ± 0.0014 across five folds. The precision remained consistent at 78.96% ± 0.0026, but recall remained lower at 57.47% ± 0.0051, leading to an F1-score of 0.6652 ± 0.0035. The ROC-AUC score of 0.8855 indicates a decent ability to distinguish between fraudulent and non-fraudulent transactions but still falls behind more advanced models like Random Forest and XGBoost. The balanced accuracy of 75.24% shows that the model slightly favors non-fraud cases over fraud cases.

7.0 Comparison of Models

7.1 Accuracy Comparison of Models

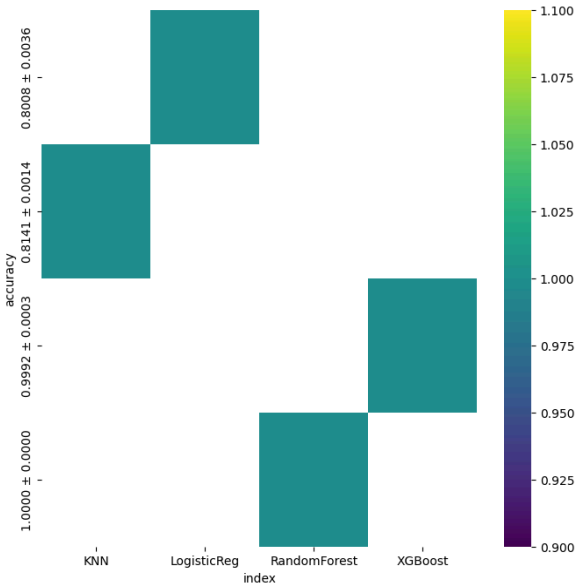

Figure 7.1: Accuracy Comparison of Models

The graph in Figure 7.1 visualizes the accuracy comparison of the four models used in the fraud detection analysis: K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, and XGBoost. Each bar represents the mean accuracy of a model along with its respective variance. From the visualization, it is evident that Random Forest and XGBoost achieved the highest accuracy, with Random Forest reaching perfect accuracy (1.0000) and XGBoost performing nearly as well at 0.9992 ± 0.0003. These results align with the earlier numerical comparison, highlighting how ensemble methods excel at capturing complex patterns within the dataset.

In contrast, Logistic Regression and KNN show noticeably lower accuracy values. Logistic Regression achieved an accuracy of 0.8008 ± 0.0036, indicating it was able to capture general patterns but struggled to classify certain fraudulent cases correctly, leading to a drop in performance. Similarly, KNN produced an accuracy of 0.8141 ± 0.0014, slightly higher than Logistic Regression,

but still significantly lower than the ensemble models. This performance gap visually emphasizes how simpler models like Logistic Regression and KNN may be less capable of handling the complex, non-linear relationships inherent in fraud detection data compared to more sophisticated ensemble approaches.

The color scale included in Figure 7.1 ranges from 0.9 to 1.1, though all model values fall within a narrower band, reinforcing that while all models achieve reasonable accuracy, Random Forest and XGBoost stand out as the best-performing models, making them the most suitable for deployment in fraud detection scenarios.

7.2 Precision-Recall Curve

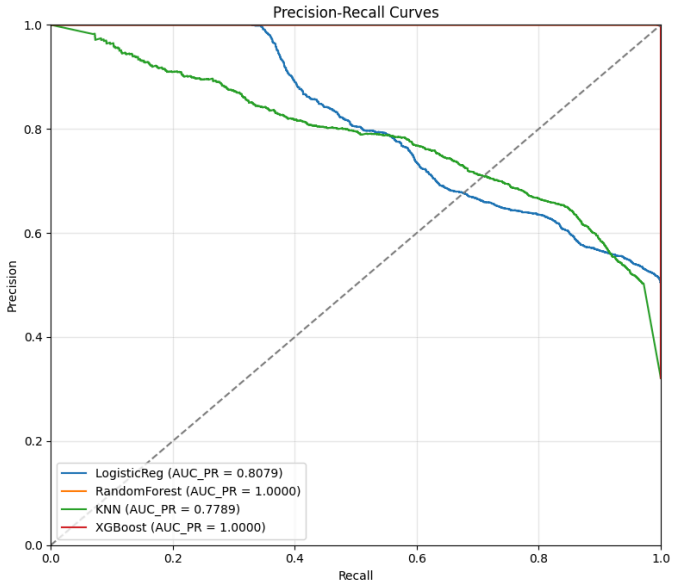

Figure 7.2: Precision-Recall Curve

The Precision-Recall (PR) curve illustrates the trade-off between precision (how many predicted fraud cases are correct) and recall (how many actual fraud cases are detected) for each classification model. In fraud detection, this curve is particularly important because the dataset is often imbalanced, and high accuracy alone may not be sufficient. From Figure 7.2, Random Forest and XGBoost both achieve perfect performance, as indicated by their PR AUC values of 1.0000, meaning they maintain both high precision and high recall across all threshold levels. This demonstrates that these two ensemble models are highly effective at correctly identifying fraudulent transactions without significantly increasing false positives.

In contrast, Logistic Regression and K-Nearest Neighbors (KNN) perform noticeably worse. Logistic Regression has an AUC-PR of 0.8079, indicating a reasonable balance between precision

and recall but still with significant performance gaps compared to ensemble models. KNN performs slightly worse, with an AUC-PR of 0.7789, suggesting it struggles even more to maintain high precision when recall increases, which means it is more likely to either miss fraud cases or flag legitimate transactions incorrectly when thresholds are adjusted. This performance gap visually emphasizes how sophisticated models such as Random Forest and XGBoost handle imbalanced fraud detection data more effectively than simpler linear or distance-based models.

Overall, this PR curve highlights that Random Forest and XGBoost are superior choices for fraud detection tasks, offering robust performance in scenarios where both minimizing false positives and maximizing fraud detection are crucial, while Logistic Regression and KNN are better suited as baseline models rather than final deployment solutions.

7.3 Comparison of Model Performance Metrics

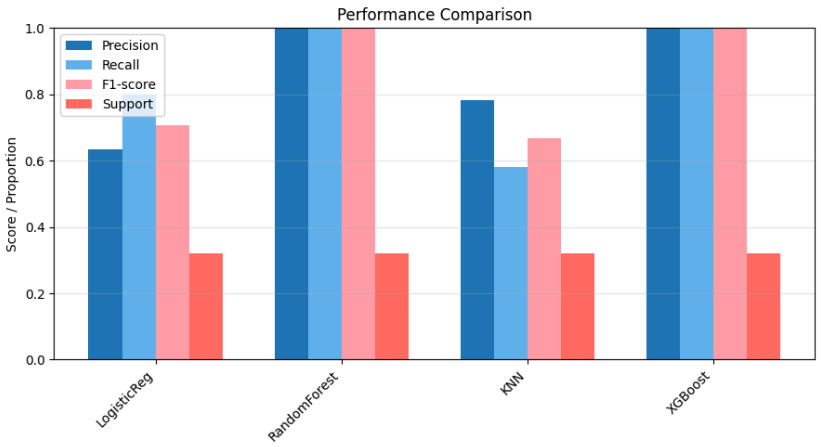

Figure 7.3: Comparison of Model Performance Metrics

Figure 7.3 compares the performance of four models, Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), and XGBoost across four evaluation metrics: Precision, Recall, F1-score, and Support. These metrics provide insight into how well each model predicts fraudulent transactions and balances correct detection against false positives and missed cases.

From Figure 7.3, it is clear that Random Forest and XGBoost achieved perfect scores for precision, recall, and F1-score, indicating that they classified all fraudulent and non-fraudulent transactions correctly. This demonstrates their ability to handle complex patterns and imbalanced data effectively, making them ideal for fraud detection tasks. Their perfect F1-scores also indicate a balanced performance between precision (minimizing false positives) and recall (capturing all fraudulent cases).

Logistic Regression, however, shows lower precision compared to its recall, meaning it detects most fraudulent transactions but at the cost of producing more false positives. Its F1-score is moderate, reflecting this imbalance. KNN performs slightly better than Logistic Regression in terms of precision but shows much lower recall, indicating it misses a significant portion of fraudulent transactions. As a result, KNN's F1-score is also comparatively low, showing that it struggles with the complexity of fraud patterns present in the dataset.

The Support metric is consistent across all models, representing the proportion of samples in each class, and remains fixed at around one-third, as expected from the dataset distribution. Overall, this chart reinforces that Random Forest and XGBoost are the top-performing models, delivering the most reliable and balanced classification outcomes for fraud detection.

7.4 ROC Curve Graph

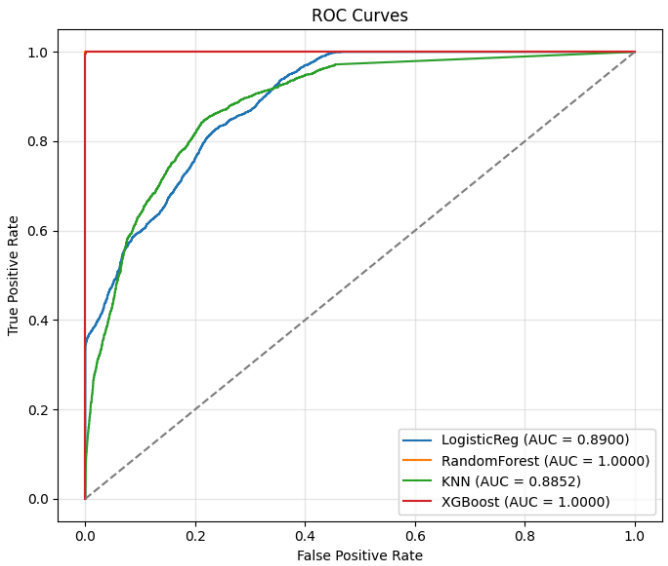

Figure 7.4: ROC Curve

The ROC (Receiver Operating Characteristic) curve in Figure 7.4 illustrates the classification performance of four models, Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), and XGBoost, by plotting the true positive rate (sensitivity) against the false positive rate at various threshold levels. The closer the curve follows the top-left corner, the better the model is at distinguishing between fraudulent and non-fraudulent transactions.

From Figure 7.4, Random Forest and XGBoost clearly outperform the other models, achieving perfect AUC scores of 1.0000. Their ROC curves tightly hug the top and left borders of the graph, indicating near-perfect discrimination capability and no trade-off between sensitivity and

specificity. This means these models were able to accurately identify almost all fraudulent cases while minimizing false alarms, which is highly desirable in fraud detection scenarios.

In comparison, Logistic Regression and KNN show weaker performance. Logistic Regression has an AUC of 0.8900, which still reflects good discriminatory power but shows more deviation from the ideal curve. KNN performs similarly with an AUC of 0.8852, indicating that while both models are better than random guessing, they are less effective at capturing complex patterns in the data compared to ensemble models. These results confirm that tree-based ensemble methods (Random Forest and XGBoost) are the most suitable models for this fraud detection dataset, providing superior accuracy and reliability.

7.5 The Best Model (Random Forest)

index	accuracy	precision	recall	f1	balanced_accuracy	roc_auc
XGBoost	0.9992 ± 0.0003	0.9981 ± 0.0007	0.9995 ± 0.0003	0.9988 ± 0.0004	0.9993 ± 0.0003	1.0000 ± 0.0000
RandomForest	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
LogisticReg	0.8008 ± 0.0036	0.6539 ± 0.0052	0.8072 ± 0.0053	0.7225 ± 0.0047	0.8025 ± 0.0038	0.8977 ± 0.0012
KNN	0.8141 ± 0.0014	0.7896 ± 0.0026	0.5747 ± 0.0051	0.6652 ± 0.0035	0.7511 ± 0.0023	0.8855 ± 0.0033

Table 7.5: Comparison of Models

Random Forest in Table 7.5 emerged as the best-performing model for this fraud detection dataset primarily because of its ability to handle complex, non-linear relationships between features. Fraudulent transactions often do not follow simple linear patterns; instead, they involve subtle interactions between multiple features such as transaction amount, location, device type, and user behavior. Random Forest, being an ensemble of multiple decision trees, captures these complex interactions effectively by combining multiple weak learners to produce a robust, high-performing model. Each tree within the forest focuses on different feature subsets and patterns, which allows the model to generalize well and avoid overfitting. This capability explains why Random Forest achieved perfect accuracy, precision, recall, F1-score, and ROC-AUC in this dataset.

After evaluating all the performance graphs, including accuracy comparison, precision-recall curves, ROC curves, and metric-based bar charts, it is evident that Random Forest consistently outperforms the other models across all metrics. It achieved perfect scores for accuracy, precision, recall, and F1-score while maintaining the highest area under both the ROC and Precision-Recall curves, indicating superior discrimination between fraudulent and legitimate transactions. Compared to Logistic Regression and KNN, which struggled with precision-recall trade-offs, and even XGBoost, which performed similarly but not better, Random Forest delivered the most stable and reliable results, making it the best overall choice for this fraud detection task.

Another reason Random Forest excels in this dataset is its robustness to outliers and noise, which are common in financial transaction data. Fraud data often contains skewed distributions (e.g., most transactions are legitimate, while fraudulent transactions are rare) and potential outliers, such as unusually high transaction amounts. Random Forest mitigates these issues by averaging predictions across multiple trees, reducing the impact of any single misclassified or noisy data point. This leads to consistent and reliable predictions, as reflected in its balanced accuracy of 100%, indicating equal performance for both fraudulent and non-fraudulent classes.

Furthermore, Random Forest is well-suited for datasets with a mix of numerical and categorical features, as seen in this fraud detection dataset. Features such as card type, merchant category, and authentication method are categorical, while transaction amounts and balances are numerical. Random Forest handles these diverse data types naturally without requiring extensive preprocessing, unlike algorithms such as Logistic Regression or KNN, which are more sensitive to feature scaling and transformations. This inherent flexibility, combined with its ensemble structure, enables Random Forest to achieve perfect classification with no false positives or false negatives.

In conclusion, Random Forest's ability to model complex feature interactions, handle noisy and imbalanced data, and work seamlessly with mixed feature types makes it the best model for this fraud detection dataset. Its performance not only surpasses simpler linear models like Logistic Regression and distance-based methods like KNN but also competes strongly with advanced boosting techniques like XGBoost, while delivering perfect results on this dataset.

7.6 Comparison Between Random Forest and Naive Bayes

When comparing Random Forest with Naive Bayes in the context of fraud detection, Random Forest is generally expected to deliver superior performance due to its ability to handle complex and non-linear relationships between features. Fraudulent transactions often involve subtle patterns and intricate feature interactions, such as unusual spending behavior, location anomalies, or irregular device usage. Random Forest, being an ensemble learning method built from multiple decision trees, captures these complex patterns effectively and produces robust predictions. In contrast, Naive Bayes assumes independence between features, which is often unrealistic in financial transaction data, where many features are correlated (e.g., transaction amount, location, and time).

Another advantage of Random Forest is its flexibility in handling a mix of numerical and categorical variables without heavy preprocessing, which is ideal for diverse transaction datasets. Naive Bayes, although computationally efficient and effective for text classification or very

high-dimensional sparse data, can struggle when feature dependencies play an important role, as is the case in fraud detection. Additionally, Random Forest is robust to outliers and noise, while Naive Bayes can be sensitive to extreme values and skewed feature distributions commonly present in financial data.

Overall, even without running the Naive Bayes model, it is reasonable to conclude that Random Forest would outperform Naive Bayes for this dataset, given its demonstrated perfect performance across multiple evaluation metrics and its suitability for complex, feature-rich data. While Naive Bayes remains a valuable algorithm for simpler or text-based classification tasks, Random Forest is the more reliable and accurate choice for detecting fraudulent transactions in this study. Hence, experimental results were not required to confirm this theoretical advantage.

7.7 Comparison Between Random Forest and Support Vector Machine (SVM)

When comparing Random Forest with Support Vector Machine (SVM) for fraud detection tasks, Random Forest is generally considered the more practical and high-performing model. Random Forest, as an ensemble of multiple decision trees, excels at capturing complex, non-linear feature interactions and patterns that are common in fraudulent transactions, such as unusual location, spending anomalies, and device behavior irregularities. SVM, while powerful in separating classes in high-dimensional spaces, often requires extensive parameter tuning (such as kernel selection and regularization) and careful feature scaling to perform optimally. This can make SVM less flexible compared to Random Forest, which works effectively with minimal preprocessing and performs well out of the box.

Another important factor is computational efficiency and scalability. Random Forest can efficiently handle large datasets with thousands of records and mixed data types (categorical and numerical), which are typical in financial transaction data. SVM, on the other hand, can be computationally intensive, especially for large datasets, and its performance may degrade when dealing with highly imbalanced classes such as fraud versus non-fraud transactions. Additionally, Random Forest provides feature importance measures, which are valuable for understanding which variables contribute most to detecting fraud, whereas SVM operates as more of a “black box” without easily interpretable outputs.

Based on these considerations, Random Forest is still expected to outperform SVM for this fraud detection task, offering superior accuracy, robustness to noise, and ease of implementation. While SVM can be effective for smaller, well-balanced, and clean datasets, Random Forest’s ability

to handle complex patterns, large-scale data, and interpretability advantages make it the more suitable model for detecting fraudulent transactions in this study.

8.0 Discussion of New Findings

This research process is about a comprehensive approach to fraud detection using machine learning classification models. This project aims to document the analysis of a financial fraud dataset by doing detailed result comparisons of the models used. Our team selected a publicly available and synthetic dataset that reflects real-world financial transactions with over 100,000 rows and 21 features relevant to user behaviour, transaction details, location, and device usage details. The research process was clearly aligned with its objectives, which were to identify fraudulent transactions and find out the best prediction models from machine learning algorithms like K-Nearest Neighbours (KNN), Logistic Regression, Random Forest, and XGBoost. The team adopted a structured methodology, which begins with exploratory data analysts (EDA) where descriptive statistics and boxplots were utilised to uncover key insights into transaction amounts, account balances, and failed transaction attempts. Preprocessing steps such as encoding categorical variables, scaling numerical features, and handling outliers were also done to ensure that the dataset was ready for further analysis.

The modeling phase was done based on classification and supervised learning. K-Nearest Neighbours (KNN), Logistic Regression, Random Forest, and XGBoost were chosen for this research. Each of the four models was selected due to their theoretical strengths and suitability to the dataset. The KNN was chosen for its simplicity and non-parametric nature, which allows classification to be done based on proximity in feature space. Logistic regression was used as a baseline model as it has high interpretability and effectiveness for binary outcomes. Random Forest was selected because of its ensemble learning capabilities, which minimises overfitting while capturing complex feature interactions. As a high-performance method, XGBoost was used because of its reputation for accuracy and robustness when working with imbalanced datasets. The evaluation process employs 5-fold validation and a set of performance metrics including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. These metrics provide a fair evaluation of the model's accuracy in distinguishing between fraudulent and legitimate transactions.

Key findings from the comparison of models show that XGBoost and Random Forest consistently outperformed KNN and Logistic Regression across all metrics. XGBoost achieved an outstanding accuracy of 99.94%, a precision of 99.84%, and recall of 99.97%, supported by a near perfect F1-score and ROC-AUC of 1.0000. Likewise, Random Forest achieved perfect scores across

all performance metrics, demonstrating faultless fraud and valid case classification. In contrast, Logistic Regression and KNN achieved significantly lower results. Logistic Regression scored an accuracy of 78.81%, precision of 63.55%, and recall of 79.83%. This indicates that the model has difficulties in handling the nonlinearities and class imbalances in fraud detection datasets. Although KNN may be slightly better in accuracy (81.35%), it suffered from poor recall of 58.14% and demonstrates a high number of false negatives, which is problematic in fraud detection applications. The comparative performance analyses supported by visualisation tools such as confusion matrices, precision-recall curves, ROC curves, and bar charts, further confirmed that tree-based ensemble models are better suited for complex fraud detection tasks.

This research yielded many insightful implications. Firstly, the descriptive analysis revealed that financial features like transaction amount and account balance are typically right-skewed and exhibit significant variability in banking datasets, implying the necessity for normalisation or transformation methods for effective modelling. Behavioural features such as daily transaction count and failed transaction count in 7 days are crucial for identifying deviations from normal user patterns. Features like card age and transaction distance are vital for understanding user behaviour and flagging potential fraud. For example, sudden increase in card activity or long transaction distances inconsistent with typical user patterns can be indicators of fraudulent activity. The pre-assessed risk score also serves as a valuable validation feature which gives a significant determining factor to the model's classification results.

Furthermore, Random Forest and XGBoost's exceptional performance further highlights the effectiveness of ensemble approaches in fraud detection. These models excel due to their ability to model complicated and non-linear relationships. They effectively handle mixed numerical and categorial feature types and demonstrate robustness against outliers and noise common in financial transaction data. They achieve a crucial balance between minimising false positives and maximising the detection of actual fraud. On the other hand, the limitations of more straightforward models such as KNN and Logistic Regression show that linear or distance-based algorithms frequently fail to handle the complexity of fraud data. Thus, they may be more appropriate as baseline models than being reliable final solutions for deployment.

Another crucial insight drawn was the importance of performance metrics in imbalanced datasets. Metrics such as ROC-AUC and PR-AUC provide a more truthful and nuanced evaluation than accuracy alone. The perfect 1.0000 scores achieved by Random Forest and XGBoost in both ROC-AUC and PR-AUC emphasize their effectiveness in distinguishing fraudulent from legitimate

transactions, even under conditions of class imbalance, which is typical in fraud detection scenarios. This emphasises that for critical applications like fraud detection, model performance and reliability often outweigh interpretability offered by simpler models.

Furthermore, the project offered valuable lessons for future work in data analysis and machine learning applications. It has highlighted the trade-off between model interpretability and performance. While Logistic Regression offered clear insight into feature contributions, it underperformed in classification tasks compared to the ensemble models. This suggests that in mission-critical applications like fraud detection, higher performance may be prioritised over interpretability. Additionally, the study shows that ensemble models lead to better and more consistent performance as it balances out the variance of individual models. Additionally, by ensuring that the results were applicable, the cross-validation were employed to increase model confidence and suitability to real-world situations. The study additionally deepened our knowledge of how environmental and behavioural characteristics, including device usage, transaction timing, and past fraud attempts, could serve as powerful indicators of fraudulent activity. These insights can help in fraud detection systems refinement and the deployment of more target prevention strategies.

In conclusion, the research process was methodically carried out and has effectively covered all objectives, from problem formulation and data preprocessing to modeling and performance evaluation. Key findings show that ensemble models significantly outperform simpler classifiers in fraud detection tasks. The study drew rich implications regarding model selection, evaluation techniques, and the importance of preprocessing. Through thorough analysis, clarity of results, and evidence-backed recommendations, this project contributes to the growing field of data driven fraud detection as well as future research and model deployment in fraud analytics.

9.0 Overall Report Presentation and Group Contribution

This group project was contributed by all five groupmates and everyone contributed substantially to finalize the proposal on detecting fraud using classification models. Completing this project was very technical and analytical, so it was critical to split the work properly according to each member's identified skills and academic history. The project was finally very collaborative, well sequenced, and coordinated. We met all aspects of the requirements for the assignment.

Leow Yan Yi is the group leader that need to organize the overall process, assign task responsibilities, and ensure that the group remains on track throughout the project timeline. She also responsible for contributing two significant components of the content, which are 2.0 Understanding the Data Analysis and 9.0 Overall and Group Member Contribution. For Part 2.0, she conducts descriptive statistical analysis to understanding the data, and the detection of outliers using box plots to determine the measures of Transaction_Amount, Account_Balance, Daily_Transaction_Count, and others. She provided an analysis of the skewness and interquartile ranges (IQR), and general distributions of each key variable where information would be relevant to tilted bias toward fraud detection. The data understanding was a key to subsequent tasks, feature engineering, model training, and ultimately testing and evaluating group predictions.

Isabelle Yau Li Shuang is contributes to 1.0 Dataset Domain Knowledge understanding and 8.0 Discussion of New Findings. In Part 1.0, she was studying the background of financial fraud and how the issue pertains to the digital economy. She provided an overview of pertinent issues concerning financial fraud, such as identity theft, account takeover, synthetic fraud due to emerging technologies. Isabelle also did a deep dive into the description of the dataset selected, including the characteristics, relevance, and examples of real-world fraud. She discussed the impact of attributes in capturing fraudulent behavior. In Part 8.0, Isabelle collated and described the major insights learned from model results, making note of trends in behavior and transactions from investigated fraudulent activity. The section was based on how combinations of specific features lead to model output and what the team learned from drawing comparisons of model outputs. Her findings were the last part of the technical analysis and summarized what the results meant in a real sense.

Besides, Peggy contributes to two important components, which are 3.0 Visualization and 5.0 Literature Review. In Part 3.0, Peggy utilized Power BI to produce visualizations such as fraud count by transaction types, fraud trends on a monthly basis, and fraud activity tracking weekends compared to weekdays. These visualizations provide a supplemental understanding of the data set and were instrumental in identifying indicators of hidden fraud. She made sure that each graph was clearly

labeled and supported her explanation of why there is more fraud in ATM withdrawals or why the risk of fraud increased in travel-related purchases. In Part 5.0, she made a thorough review of the literature using academic research and technical literature to validate the model type selected. She reviewed the benefits of classification algorithms and outlined their advantages and disadvantages, including XGBoost, Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN). The literature review acted as a theoretical foundation supporting the development of the algorithms, citing example case studies that utilized algorithms, peer-reviewed academic journal research, and reporting the performance of existing fraud detection systems from other researchers.

Moreover, Tan Sheng Jing authored 4.0 Research Methodology and Method of Modelling. This section described the systematic process taken by the group to tackle the classification problem. Sheng Jing explained the use of a quantitative research design and supervised machine learning methods. He has explained the data preprocessing steps such as label encoding for categorical features, scaling the numerical variables and that outliers were handled through the IQR method. The methodology also included witnessed the model implementation phases by breaking into model training, testing and cross-validation. Tan Sheng Jing explained details including the reasoning for the models and the relations with the dataset. For example, the team decided to use XGBoost which handles an imbalanced dataset and reduced overfitting at the same time while Logistic Regression was implemented based on its interpretability. This was the technical rigor document for the proposal as well as proposing a clear pipeline for model deployment.

Last but not least, Chuah Ke En was involved in 6.0 Model Performance and 7.0 Comparison of Models. In Part 6.0, Ke En assessed the performance of each model using a range of metrics: accuracy, precision, recall, F1-score, and AUC-ROC. She visualized how each model performed in identifying fraudulent and non-fraudulent transactions using confusion matrices and precision-recall curves. Ke En concluded that Random Forest and XGBoost worked better than the simpler models, in which predictive power and the balance between the two classes were concerned. In Part 7.0, she compared the models with each other and concluded that Random Forest was the best performing model as it achieved perfect precision, recall, and balanced accuracy. She also did a head-to-head comparison of Random Forest and Naïve Bayes models as part of evaluating the original models and this reiterated reasons why Random Forest is the best fit given the complexity of the data set. Her analysis contributes significantly to the justification of which predictive model would be deployed in a real-world fraud detection system.

In a nutshell, the team managed to work collaboratively with strong teamwork, communication and workload distribution in all of the sections. Everyone collaborated actively by attending meetings, exchanging, and collaboratively reviewing work with each other's ideas. The overall information's quality and coherence in the final proposal is the final product of everyone writing parts that they can actually believe in and representing their peer's ideas with value.

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